Annexure 03

General Sir John Kotelawala Defence University
Faculty of Computing
IT42999 - Individual Research Project

APPLICATION FOR UNDERGRADUATE FINAL YEAR RESEARCH PROJECT

1. Project Title *The title of the project should be brief, reflect concisely and accurately the proposed. Students are advised to avoid titles which convey a distant or potential application of the proposed work, or a greater aspiration or goal than is to be expected from the proposed work.*	
Smart Career Pathway and Learning Recommendation System	
2. Research Area	**Specific Area**
Artificial Intelligence and Machine Learning	Educational Technology Personalized Learning Systems AI-Based Course Recommendation for Student Skill Development Step-by-Step External Learning Path Recommendation
3. Supervisor(s)	
Name and Designation: Ms. LDTT De Silva Institution (if not from the University): Area of expertise related to the proposed project:	Mobile :+94 76 963 5871 E-mail : thumashids@kdu.ac.lk
Name and Designation: Institution (if not from the University): Area of expertise related to the proposed project:	Mobile E-mail
Name and Designation: Institution (if not from the University): Area of expertise related to the proposed project:	Mobile E-mail
4. STUDENT	
Name: M.H.M.H. Hiruvinda Index No: D/BIT/23/0014	Mobile : +94 70 192 0762 E-mail : 40-bit-0014@kdu.ac.lk

5. Summary

(a) Explain briefly the research problem, research approach and expected outputs.
Not exceeding one A4 page, font type Times New Roman, size 11, single space.

Selecting suitable learning paths and skill development opportunities is a major challenge faced by students, especially at early stages of education. Many students lack proper guidance to identify their interests, strengths, and appropriate learning levels, resulting in unstructured learning and ineffective skill development. Existing online learning platforms primarily focus on listing courses and do not provide personalized, step-by-step learning guidance. Furthermore, students often struggle to determine which beginner-level resources to start with and how to progress toward advanced skill levels in a structured manner.

The proposed research aims to design and develop an AI-based personalized learning recommendation system to support student skill development. The research adopts a design-and-development approach, focusing on the implementation of a student-centered web application as a learning guidance tool. The system collects student profile data, including interests, education level, talents, and learning preferences, and analyzes this data using Artificial Intelligence and Machine Learning techniques to determine the student's current learning level and suitable learning path.

Based on the analysis results, the system recommends online learning resources in a structured progression from beginner to intermediate and advanced levels. The system does not host or manage learning content but provides curated recommendations to platforms such as online courses and tutorials. Student learning progress is tracked by allowing users to mark completed courses, enabling the system to dynamically update future recommendations. This approach ensures continuous and personalized learning guidance tailored to individual student needs.

The expected outputs of this research include a functional AI-based learning recommendation application, a personalized course recommendation mechanism, a learning progress tracking module, and structured learning pathway generation. The study contributes to the field of educational technology by introducing a student-focused intelligent learning guidance system that supports informed decision-making and effective skill development through personalized recommendations.

(b) Give 3 – 5 keywords for the proposed project:

Artificial Intelligence, Personalized Learning, Course Recommendation System, Student Skill Development, Educational Technology

6. Research Problem

6.1 Research problem/s

Students often face difficulties in choosing the right career path due to limited guidance, lack of awareness of their strengths, and a rapidly changing job market. In Sri Lanka, career counseling resources are scarce, and existing platforms provide generic suggestions without considering individual talents, habits, or preferences. As a result, students may pursue unsuitable courses or careers, leading to underperformance and frustration.
Similarly, companies struggle to find candidates whose skills and interests match job requirements, as there is no integrated system connecting student skills with career opportunities. Consequently, both students and employers lack personalized guidance and effective matching, highlighting the need for an intelligent system that provides tailored career paths, learning recommendations, and job matching.

6.2 Analysis of the problem/s & rationale for the research question

Research shows that many students struggle to identify suitable career paths due to limited guidance, lack of awareness of their strengths, and rapidly changing job markets. Existing platforms provide generic recommendations without considering individual talents, habits, or learning behavior. This often leads students to choose unsuitable courses or careers, causing frustration and underperformance.

Similarly, companies face difficulties finding candidates whose skills and interests match job requirements, as there is no system that connects students' abilities with career opportunities.

This research is motivated by the need for an intelligent system that analyzes students' preferences, talents, and learning behavior, and provides personalized career paths, learning recommendations, and job matching.

The research question is therefore:
How can a web-based intelligent system analyze students' skills and preferences to recommend suitable career paths, learning resources, and job opportunities?

7. Comprehensive literature review AND the complete list of references in the relevant area.
(Attach additional sheets if necessary)

1. AI-Driven Career Recommendation Systems
AI-driven career recommendation systems use machine learning to analyze student data and suggest personalized career paths. Studies show that models such as RNNs with attention mechanisms can effectively match students to suitable careers based on their academic performance and personal interests [1]. Similarly, systems using random forests and neural networks leverage real-time job and course data to recommend career opportunities, providing dynamic and context-aware suggestions [2].
These approaches improve over traditional guidance by providing personalized and data-driven recommendations, reducing reliance on generic counseling. However, they often require structured input data and may not integrate psychometric or behavioral insights into their recommendations [3].

2. Personalized Career Guidance
Personalized career guidance systems aim to combine multiple sources of student information, such as skills, habits, and preferences, to offer more accurate recommendations. ML-based systems that include explanation features help students understand why a particular career or course is suggested, improving transparency and trust in the recommendations [4]. Hybrid approaches that integrate collaborative and content-based filtering have been shown to further enhance personalization by considering both student similarity and career attributes [5].
Integrating gamification and psychometric assessments can additionally increase engagement and motivation among students, guiding them through learning pathways and career exploration in an interactive manner [6].

3. Hybrid and Fuzzy Recommendation Models
Hybrid and fuzzy recommendation systems combine multiple techniques to handle the uncertainty and variability in student preferences. Fuzzy logic-based systems can accommodate ambiguous or imprecise student input, while hybrid systems leverage both academic and personal features to improve recommendation accuracy [7]. Such systems demonstrate better adaptability to diverse student profiles and provide more flexible career guidance.

4. Learning Path and Course Recommendations

To support long-term career goals, learning path recommendation systems suggest specific courses, skills, and training aligned with a student's strengths and interests. ML-based models can recommend both domain-specific courses and broader learning pathways, while collaborative filtering helps identify courses pursued by similar students [8]. Additionally, integrating skill-gap analysis allows students to focus on areas requiring improvement, aligning education with career requirements [9].
By linking career recommendations with personalized learning paths, these systems aim to enhance employability and ensure that students follow actionable steps toward their goals [10].

References

[1] D. Kiranmai, T. Sumanth, G. Chandrika, B. U. Varma, and S. Ishaq, "AI Driven Career Path Recommendation System," vol. 13, no. 2, 2025.

[2] E. S. Vignesh, V. Sanjay, and M. A. Akileshwari, "AI-Based Career Path Recommendation System," vol. 11, no. 10, 2025.

[3] V. Maphosa and M. Maphosa, "Fifteen Years of Recommender Systems Research in Higher Education: Current Trends and Future Direction," *Appl. Artif. Intell.*, vol. 37, no. 1, p. 2175106, Dec. 2023, doi: 10.1080/08839514.2023.2175106.

[4] P. Raj. S. Goutham Krishna, A. M. Bharath G, and A. V. D. Neelam Sanjeev Kumar, "PERSONALIZED LEARNING PATH RECOMMENDATION SYSTEM," *IJIREEICE*, vol. 13, no. 11, Nov. 2025, doi: 10.17148/IJIREEICE.2025.131102.

[5] M. Kumar, A. Raj, and S. Kumar, "Career Counselling Recommendation System," Apr. 18, 2025, *Computer Science and Mathematics*. doi: 10.20944/preprints202504.1542.v1.

[6] S. Gowsika, S. Aparna, V. Kiruthika, and K. Lavanya, "Personalized Career Guidance System for Students," *Int. J. Adv. Res. Sci. Commun. Technol.*, pp. 27–32, Mar. 2025, doi: 10.48175/IJARSCT-23904.

[7] X. Q. Ong and K. H. Lim, "SkillRec: A Data-Driven Approach to Job Skill Recommendation for Career Insights," Feb. 20, 2023, *arXiv*: arXiv:2302.09938. doi: 10.48550/arXiv.2302.09938.

[8] R. Soni, B. Suleiman, and S. Singh, "Improving Graduate Outcomes by Identifying Skills Gaps and Recommending Courses Based on Career Interests," Nov. 12, 2025, *arXiv*: arXiv:2511.09819. doi: 10.48550/arXiv.2511.09819.

[9] "[No title found]," *Int. Res. J. Mod. Eng. Technol. Sci.*.

[10] D. Thakur, R. Rathod, D. Divekar, S. Deshmukh, and P. Chavan, "Survey paper-AI powered career recommendation for students," vol. 9, no. 6, 2025.

8. Originality & innovativeness of the proposed work

The originality of the proposed work lies in its focus on delivering an AI-driven career guidance and learning recommendation system tailored to the educational and employment context. While many existing career guidance platforms provide generalized recommendations, they often do not consider individual preferences, talents, learning behavior, or local job market requirements. The proposed system addresses this gap by offering personalized career pathways aligned with users' educational backgrounds, interests, and skill development needs.

Another innovative aspect of this project is the integration of machine learning-based user profiling with learning and job recommendation mechanisms. Instead of providing static or rule-based career suggestions, the system analyzes user attributes and continuously updates recommendations as users acquire new skills or change preferences. This adaptive approach enables more accurate and dynamic career guidance compared to traditional counseling systems.

Furthermore, the system uniquely combines career pathway recommendation with personalized learning resource suggestions, including both free and paid courses, and job matching functionalities within a single platform. By linking career goals to actionable learning paths and real-world

employment opportunities, the proposed work enhances user engagement and practical career readiness. Overall, the integration of intelligent career analysis, adaptive learning recommendations, and job-market alignment within one platform represents a novel and innovative contribution to AI-based career guidance systems in the context.

9. Overall aim and specific objectives of the proposed work

9.1 Overall aim

The overall aim of this project is to design and develop an AI-driven career pathway and learning recommendation system that analyzes users' interests, skills, and educational background to provide personalized career guidance, learning pathways, and job matching opportunities.

9.2 Specific Objective/s

- To study common challenges faced by students in identifying suitable career paths based on their interests, skills, and educational background.
- To design and develop an intelligent web based career guidance system that supports personalized career pathway planning.
- To implement a machine learning based user profiling mechanism using data such as academic qualifications, interests, talents, and learning preferences.
- To develop a personalized learning and career recommendation module that suggests suitable career paths, courses (free and paid), and skill development activities.
- To provide job matching and progress-tracking features that allow users to monitor skill development and career readiness.

10. Methodology

10.1 Describe the Methodology
(*Attach additional sheets if necessary)*

The methodology for developing the Smart Career Pathway and Learning Recommendation System involves several key stages: Research and Analysis, System Design, Development and Implementation, Testing and Validation, and Deployment. The research methodology combines qualitative and quantitative approaches, including literature review, surveys/interviews with students and employers, and prototype testing. Below is the detailed methodology for the project.

Research and Analysis

- Literature Review: Conduct a comprehensive review of existing career guidance systems, AI-powered learning recommendation platforms, and job matching technologies used locally and internationally.
- Problem Identification: Identify limitations in current career guidance approaches, including lack of personalized recommendations, fragmented resources, and limited alignment with real job market requirements.
- Stakeholder Surveys/Interviews: Engage students, graduates, and employers to understand challenges in career planning, skills gaps, course selection, and job matching preferences.

System Design

- User Requirement Specification: Define functional and non-functional requirements for students, employers, and administrators based on stakeholder insights.
- Interface Design: Design a user-friendly web interface following Human-Computer Interaction (HCI) principles to simplify profile creation, course submission, and career guidance.
- Architecture Design: Develop the technical architecture including frontend (React.js), backend (FastAPI), machine learning modules, hybrid recommendation engine, knowledge graph (Neo4j), and database storage (MongoDB).

- Integration Planning: Plan integration of core functionalities including user profiling, ML-based career recommendation, course completion tracking, job matching, and skill progress monitoring.

Development and Implementation

- Platform Development: Build the web application using React.js for the frontend and FastAPI for the backend, ensuring scalability and responsiveness.
- Recommendation Engine: Implement a hybrid recommendation system combining ML models (Random Forest, KNN) and NLP embeddings (Sentence-BERT) to suggest personalized career paths and learning resources.
- Course Completion Tracking: Enable users to upload course certificates (image/PDF), which are stored in MongoDB/GridFS, updating skill profiles and unlocking subsequent career guidance steps.
- Job Matching Module: Compare user skills with job requirements to suggest suitable employment opportunities.

Testing and Validation

- Unit Testing: Test individual components (profile creation, recommendation engine, course submission) for correctness.
- User Testing: Conduct usability testing with students and graduates to ensure the system is intuitive and supports effective career guidance.
- Performance Testing: Evaluate system performance under concurrent user load, ensuring fast recommendation response times.
- Security Testing: Implement authentication, secure storage of user data and course certificates, and ensure compliance with data privacy standards

10.2 Experimental design where applicable
Please complete relevant sections

Functional Requirements

1. User Registration and Login
• Student register and log in
• Each user shall have a secure and authenticated account.

2. User Profile Management
• Update personal details such as name, age, and education level
• Add and edit interests, talents, and skills

3. Preference and Skill Input
The system shall allow users to complete questionnaires or forms to collect:
• Interests , Talents , Habits , Career preferences

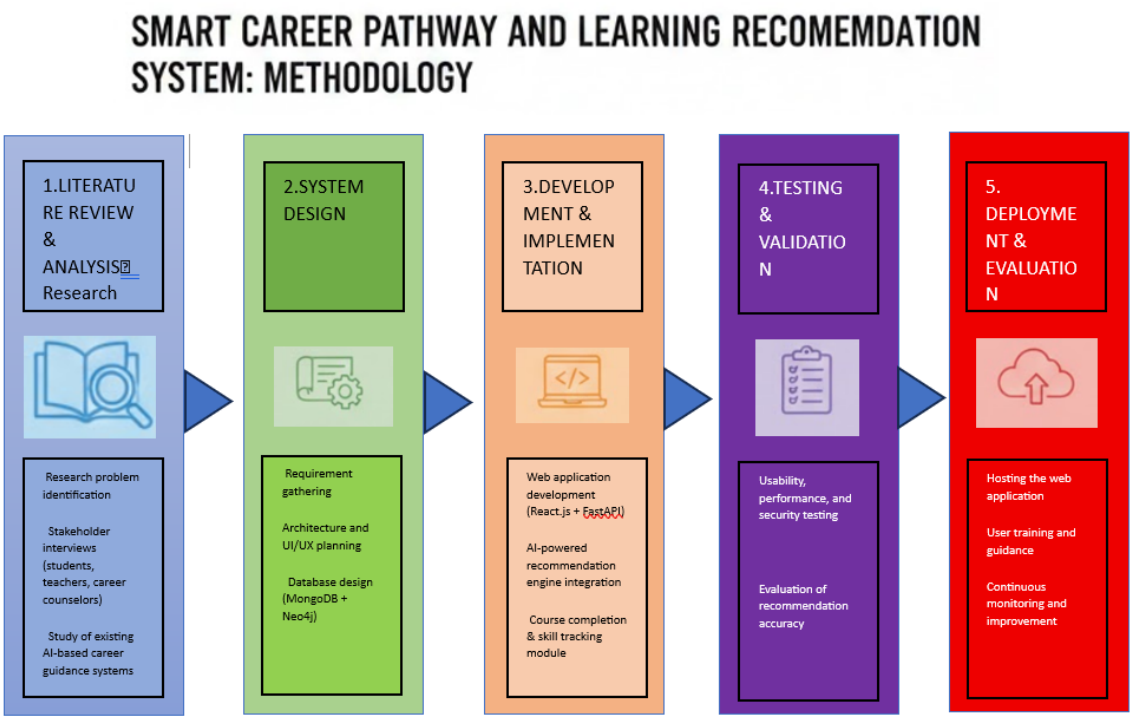
4. Career Path Recommendation
The system shall:
• Generate personalized career pathways
• Provide a step-by-step learning and career roadmap
• The roadmap shall be tailored to the user's age and education level.

5. Course recommendation
The system shall suggest:
• Free online learning resources
• Paid professional courses
• Courses shall be recommended in a structured order: beginner → intermediate → advanced.

6. Progress Tracking
The system shall allow users to:
• Mark recommended courses as completed
• Update skills and qualifications
• The system shall automatically update future recommendations based on user progress.

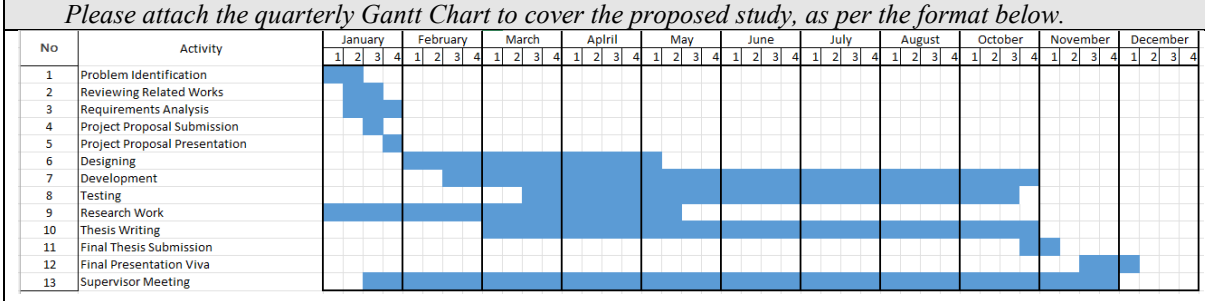
Non-Functional Requirements

- Performance - The system shall respond quickly to user actions , Recommendations shall be generated within a few seconds.
- Scalability - A large number of student users
- Security - Securely store all user data , Encrypt user passwords , Prevent unauthorized access
- Reliability - The system shall be available 24/7 , Regular backups shall be maintained to prevent data loss.



10.3 **Work plan**

Please attach the quarterly Gantt Chart to cover the proposed study, as per the format below.



10.4 **Indicators & milestones of progress**

Please list the milestones and indicators that will be used to measure the progress of the proposed Study

The progress of the proposed study will be monitored through clearly defined project milestones and measurable indicators. Each milestone represents a key development phase of the system, while the indicators are used to assess the successful completion of each phase.

1. Literature Review and Requirement Analysis
   - Completion of a comprehensive literature review related to AI-powered career guidance, machine learning-based recommendation systems, and student skill mapping.
   - Documentation of functional and non-functional system requirements based on student and employer surveys/interviews.
2. System Design
   - Completion of system architecture design, including frontend (React.js), backend (FastAPI), recommendation engine, and database schema (MongoDB + Neo4j).
   - Finalization of user interface designs for students, employers, and administrators, ensuring usability and accessibility.
3. Development of User Profile and Career Tracking Module
   - Implementation of user registration, profile creation, and skill tracking features.
   - Successful capture of user data, including interests, educational background, completed courses, and career goals.
4. Implementation of ML-Based Recommendation Engine
   - Development of a hybrid recommendation system combining ML models (Random Forest/KNN), NLP embeddings (Sentence-BERT), and collaborative filtering.
   - Validation of recommendation outputs through test cases and sample profiles.
5. Course Completion and Skill Update Module
   - Implementation of course certificate upload functionality and integration with MongoDB/GridFS.
   - Automatic update of user skills based on verified course completion.
6. Job Matching and Career Pathway Module
   - Implementation of job matching based on updated skills and career recommendations.
   - Integration with Neo4j knowledge graph for career pathway visualization.
7. System Integration and Testing
   - Successful integration of all modules: user profile, recommendation engine, course tracking, and job matching.
   - Functional testing, bug fixing, and usability testing completed.
8. Evaluation and Data Analysis
   - Collection and analysis of system-generated data to assess accuracy of career recommendations, course suggestions, and skill progression.
   - Feedback obtained from students and employers to evaluate usability, relevance, and effectiveness.
9. Final Documentation and Submission
   - Completion of final project report, system documentation, and technical manuals.
   - Submission of the fully functional web application along with research findings.

11. Ethical considerations

Relevance to the project	Relevant ☒ Not relevant ☐

12. Plan to protect the human & environmental safety issues related to the project

Human Safety and Ethical Considerations

- Participation of students and recent graduates will be voluntary, with informed consent obtained during system registration.
- Any testing or feedback sessions will be conducted under supervision of academic advisors or project supervisors.

- The system will serve solely as a career guidance and learning support tool and will not provide professional career counseling or legal advice.
- User interfaces and content will be designed to be age-appropriate, inclusive, and non-biased, ensuring a positive experience.
- Interaction tasks (profile updates, course submissions, and recommendations) will be structured to avoid fatigue or information overload.

Data Privacy and Security

- Only essential user data related to career interests, skills, and learning progress will be collected.
- Personal data will be anonymized and securely stored in MongoDB, with access restricted to authorized users and system administrators.
- Certificates and sensitive documents will be stored using secure cloud storage with proper access controls.
- Data collected will be used solely for system functionality, academic research, and improvement of recommendations.
- Development and testing will follow best practices to ensure data protection and compliance with privacy regulations.

Environmental Safety

- The project is entirely software-based, with no hazardous materials or physical equipment involved.
- By providing digital career guidance and course tracking, the system reduces reliance on printed career handbooks and materials, minimizing environmental impact.
- Development and testing will be conducted using existing computing resources efficiently to reduce energy consumption.

13. Expected Research Outputs

Deliverables at the end of the project in point form

- A fully functional web-based career guidance and learning recommendation system for students and recent graduates.
- A hybrid AI-powered recommendation engine that suggests personalized career pathways, courses, and job opportunities.
- A course completion and skill tracking module that updates user profiles based on verified certificates.
- A job matching module that recommends employment opportunities based on updated skills and career goals.
- User interface designs and system architecture documentation, including frontend, backend, and database modules.
- Evaluation results and analysis of recommendation accuracy, usability, and user satisfaction from surveys and system data.
- A final project report documenting the research, system design, implementation, evaluation, and findings.

14. Plan to utilize the research output

- The developed web application can be used by students and recent graduates as a personalized career guidance and learning support tool.
- Educators and career counselors can utilize the recommendation insights and skill tracking reports to advise students more effectively and identify skill gaps.

- Users can leverage the system to plan their learning paths, complete relevant courses, and track skill development to achieve career goals.
- The platform can be expanded to include institutional integration, allowing universities or training centers to provide tailored guidance for larger student cohorts.
- The research findings, AI recommendation engine, and system design can serve as a foundation for future research in AI-driven career guidance, educational technology, and skill development analytics.

15. Research Outcomes

15.1 Significance of research outcomes and the impact on national/socio-economic development of Sri Lanka.
Please write in point form.

- Contributes to skill development and career readiness by providing a personalized AI-based career guidance system for students and graduates.
- Supports informed decision-making for students, helping reduce mismatched career choices and unemployment rates.
- Enhances the effectiveness of career counselors and educators by offering data-driven insights into student skills and learning preferences.
- Empowers students to actively plan their learning paths, complete relevant courses, and track skill development, promoting self-directed learning and lifelong skill acquisition.
- Promotes the use of affordable web-based technology, increasing accessibility for students .
- Lays the foundation for future research and innovation in AI-powered educational and career guidance systems .

15.2 Identify relevant stakeholders and potential beneficiaries

- Students : Individuals using the platform for personalized career guidance, learning path recommendations, and skill development tracking.
- Educational Institutions / Universities: Schools, colleges, and training centers that can integrate the system to support student career planning and guidance.
- Career Counselors and Educators: Professionals using the system to provide tailored advice and monitor student progress.
- Technology Developers / IT Team: Engineers and developers responsible for building, maintaining, and updating the web application and AI recommendation engine.
- Government : Authorities overseeing education, youth skill development, and ICT initiatives in Sri Lanka.
- Academic and Research Stakeholders: Researchers in educational technology, AI-driven guidance systems, and skill development analytics.

16. Plan to protect and exploit Intellectual Property (IP)?
(Indicate if applicable)

IP Protection:

- All libraries, tools, and resources used will be properly acknowledged.
- User data, preferences, and career profiles will be kept private and used only for academic purposes.
- The system and reports will include copyright notices where applicable.

IP Exploitation:

- The application can be further developed as a commercial career guidance and job-matching platform.
- The research and system design can be used for future studies or AI/ML enhancements.

- Unique features like AI-based career recommendations and job-user matching can be patented or trademarked.

17. Signature of Student and Supervisor(s)

The details given above are correct and personally verified. **(a)** …… (Student)	…18/01/2026…… Date
(b) …… (Principal Supervisor)	…18/01/2026…….. Date
(c) …… (Co- Supervisor)	……..... Date

18. Recommendation and Approval

The application is recommended and approved.

……...…… Department Project Coordinator	…….. Date
……...…… Head of the Department	…….. Date

Guidelines for Applicants

1. The proposed research project should necessarily be an original investigation.
2. The problem to be tackled should be clearly identified.
3. A comprehensive literature review should be done, and all details should be provided together with the list of references.
4. The budget should be calculated correctly and stated in detail.
5. A detailed time-based work plan (Gantt chart) should be included stating the proposed activities and time frames clearly.
6. In the case of selection of a supervisor from an outside institution, it is mandatory to get an internal supervisor appointed.
7. Softcopy of the application should be submitted as PDF format.
8. Incomplete applications **will not** be considered.